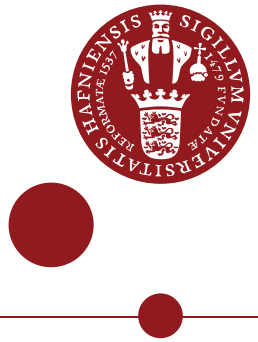
UNIVERSITY OF COPENHAGEN



Quantum Information Theory

Saving and Breaking Entanglement

Project Outside the Course Scope

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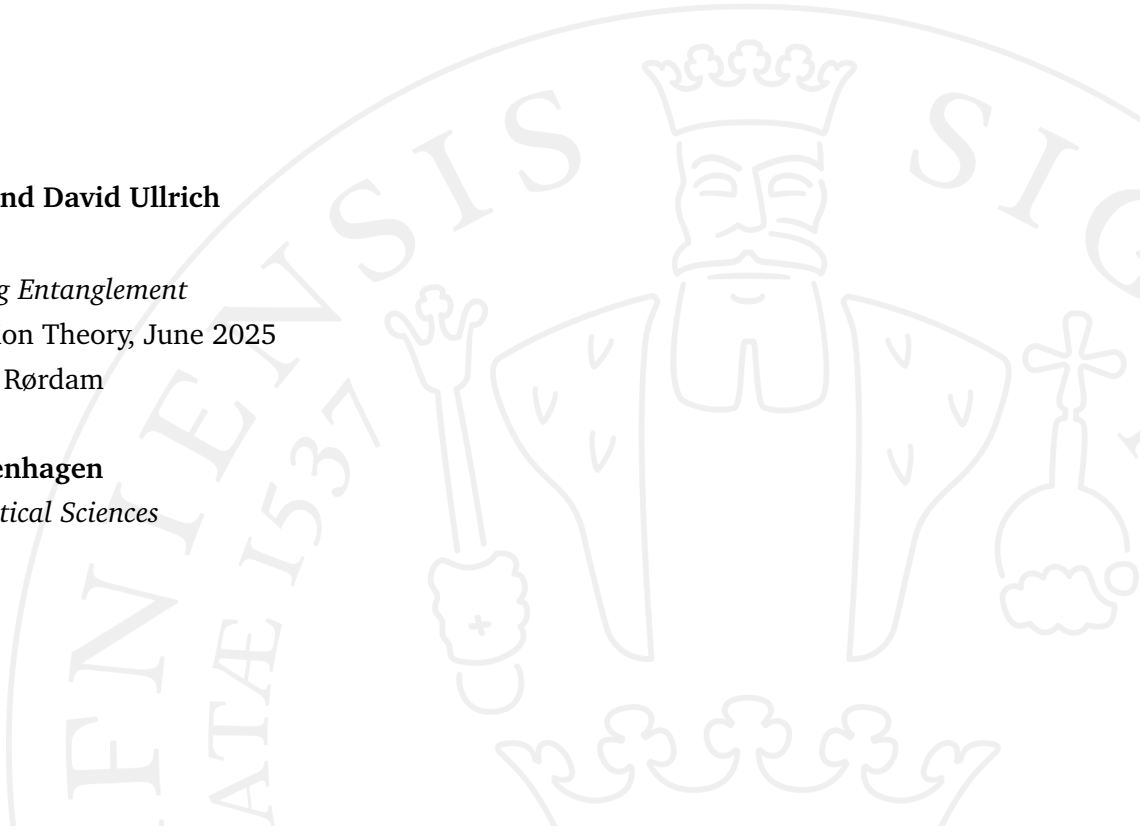
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Saving and Breaking Entanglement
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Abstract

This report explores the mathematical structure and dynamical behavior of entanglement-breaking (EB) and entanglement-saving (ES) quantum channels, with a particular emphasis on the PPT squared conjecture and the spectral classification of universally entanglement-preserving (UEP) channels. We provide a rigorous treatment of the relevant background in quantum information theory, including complete positivity, separability criteria, and spectral properties of quantum maps. Building on recent results, we analyze the structural conditions under which a quantum channel preserves entanglement indefinitely. In addition to the theoretical framework, we implement numerical simulations to investigate the entanglement-breaking index of random quantum channels, confirming the extreme rarity of ES channels and the asymptotic entanglement-destroying effect of generic noise. Our findings support the conjectural picture that entanglement-saving dynamics are exceptional, and that most noise accumulates irreversibly toward classicality.

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1 Introduction

Entanglement is one of the most fundamental features of quantum mechanics and a key resource for quantum information processing. However, entanglement is fragile and can be destroyed by the action of noise, typically modeled as quantum channels—completely positive, trace-preserving (CPT) maps acting on quantum states. Understanding when and how entanglement is lost under repeated application of a quantum channel is crucial for both the theoretical foundations and practical implementation of quantum technologies.

A central concept in this context is that of entanglement-breaking (EB) channels, which, when applied locally, always yield a separable output state, regardless of the initial entanglement. More nuanced is the notion of entanglement-saving (ES) channels, which never become entanglement-breaking, no matter how many times they are applied. The rarest among these are the universal entanglement-preserving (UEP) channels, which preserve entanglement for all input states and all ancilla systems—they turn out to be exactly the unitary evolutions.

A related and still open problem is the PPT squared conjecture, which posits that the composition of two completely positive and completely copositive maps is always entanglement-breaking. This conjecture is deeply connected to the structure of quantum noise and the geometry of separable states.

In this report, we rigorously explore these concepts, combining theoretical results, detailed proofs, and numerical experiments. Our aim is to provide a self-contained, mathematically precise, and physically motivated account of the landscape of entanglement-breaking, entanglement-saving, and universal entanglement-preserving channels, with a focus on the connections to the PPT squared conjecture.

2 Preliminaries and Mathematical Background

In this chapter, we introduce the essential definitions and results used throughout the rest of this project. The discussion is restricted to finite-dimensional Hilbert spaces, and we adopt standard notation and conventions from quantum information theory.

Quantum States and Density Matrices

A *quantum state* is described by a density matrix ρ, which is a positive semidefinite operator with unit trace acting on a complex Hilbert space $\mathcal{H} \cong \mathbb{C}^d$. That is,

$$\rho \in \mathcal{D}_d := \{\rho \in \mathcal{M}_d(\mathbb{C}) \,|\, \rho \geq 0,\ \mathrm{Tr}(\rho) = 1\}.$$

A state $\rho \in \mathcal{D}_{AB}$ on a bipartite system $A \otimes B$ is said to be *separable* if it can be written as

$$\rho = \sum_i p_i \, \rho_A^{(i)} \otimes \rho_B^{(i)},$$

where $\rho_A^{(i)}$ and $\rho_B^{(i)}$ are local density matrices and $p_i \geq 0$, $\sum_i p_i = 1$. We denote the set of separable states by $\mathrm{Sep}(d_A, d_B)$. It is convex and closed.

A state that is not separable is called *entangled*.

Bounded Linear Operators

Let $\mathcal{H}$ be a (complex) Hilbert space. We denote by $\mathcal{B}(\mathcal{H})$ the set of all bounded linear operators on $\mathcal{H}$, that is,

$$\mathcal{B}(\mathcal{H}) := \{T : \mathcal{H} \to \mathcal{H} \text{ linear and bounded}\}.$$

An operator T is called *bounded* if there exists a constant $C > 0$ such that

$$\|T\psi\| \leq C\|\psi\| \quad \text{for all } \psi \in \mathcal{H}.$$

In finite-dimensional quantum information theory, we usually assume that $\mathcal{H} \simeq \mathbb{C}^d$, in which case $\mathcal{B}(\mathcal{H})$ can be identified with the space $\mathcal{M}_d$ of $d \times d$ complex matrices.

Quantum Channels and Superoperators

A *quantum channel* (or superoperator) is a completely positive and trace-preserving (CPT) linear map $\Phi : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_d(\mathbb{C})$.

Definition 2.1. A map $\Phi : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_d(\mathbb{C})$ is:

- *Positive (P)* if $\Phi(X) \geq 0$ whenever $X \geq 0$.
- *Completely positive (CP)* if $I_n \otimes \Phi$ is positive for all $n \in \mathbb{N}$.
- *Trace-preserving (T)* if $\mathrm{Tr}[\Phi(X)] = \mathrm{Tr}[X]$ for all X.
- *Unital (U)* if $\Phi(\mathbb{1}) = \mathbb{1}$.

The Kraus representation theorem states that any CPT map Φ admits a decomposition

$$\Phi(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \text{with} \quad \sum_k K_k^\dagger K_k = \mathbb{1}.$$

Definition 2.2. For a linear transformation $Y : \mathbb{C}^{d_1} \to \mathbb{C}^{d_2}$, we define the corresponding completely positive map

$$\mathrm{Ad}_Y : \mathcal{M}_{d_1} \to \mathcal{M}_{d_2}$$

as the completely positive map given by

$$\mathrm{Ad}_Y(X) := YXY^\dagger, \quad \text{for all } X \in \mathcal{M}_{d_1}.$$

Definition 2.3. The space of linear maps from $\mathcal{H}_1 \to \mathcal{H}_2$ is denoted by $\mathrm{L}(\mathrm{H}_1, \mathcal{H}_2)$

Maximally Entangled States and the Choi Representation

Definition 2.4. The (unnormalized) maximally entangled state in $\mathbb{C}^d \otimes \mathbb{C}^d$ is defined as

$$|\Omega_d\rangle = \sum_{i=1}^{d} |i\rangle \otimes |i\rangle.$$

and its corresponding density matrix is $\omega_d = |\Omega_d\rangle \langle\Omega_d|$.

Definition 2.5. The Choi matrix of a quantum channel Φ is given by

$$J(\Phi) = (\Phi \otimes I)(|\Omega\rangle\langle\Omega|).$$

By Choi's theorem, Φ is completely positive if and only if $J(\Phi) \geq 0$.

Tricks with the maximally entangled state

1) Any vector $|\psi\rangle \in \mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$ can be written as

$$|\psi\rangle = (\mathbb{I}_{d_1} \otimes A)\,|\Omega_{d_1}\rangle = (B \otimes \mathbb{I}_{d_2})\,|\Omega_{d_2}\rangle,$$

for some linear maps $A : \mathbb{C}^{d_1} \to \mathbb{C}^{d_2}$ and $B : \mathbb{C}^{d_2} \to \mathbb{C}^{d_1}$.

2) For any Hermiticity-preserving linear map $L : \mathcal{M}_{d_1} \to \mathcal{M}_{d_2}$, we have the identity:

$$(\mathrm{id}_{d_1} \otimes L)(\omega_{d_1}) = (\vartheta_{d_1} \circ L^* \circ \vartheta_{d_2} \otimes \mathrm{id}_{d_2})(\omega_{d_2}),$$

where L^* denotes the Hilbert–Schmidt adjoint of L, and ϑ_d denotes the transpose map on $\mathcal{M}_d$.

Entanglement-Breaking and Entanglement-Saving Channels

Definition 2.6. A channel $\Phi \in \mathbf{CPT}_d$ is *entanglement-breaking (EB)* if for every ancillary space $\mathbb{C}^n$ and any bipartite input state $\rho_{AB} \in \mathcal{D}_{dn}$,

$$(\Phi \otimes I_n)(\rho_{AB}) \in \mathrm{Sep}(d, n).$$

A very important property that characterizes EB channels is that their Choi matrix is separable.

Definition 2.7. The *entanglement-breaking index* of a channel Φ is

$$n(\Phi) := \min\{n \in \mathbb{N} \mid \Phi^n \in \mathrm{EB}\} \in \mathbb{N} \cup \{\infty\}.$$

Definition 2.8. A channel Φ is *eventually entanglement-breaking (EEB)* if $n(\Phi) < \infty$.

A useful property we will use is:

Proposition 2.1. *The composition of an EB channel with any other channel is again EB.*

Partial Transpose and the PPT Criterion

Let ρ_{AB} be a bipartite density matrix. Its partial transpose with respect to subsystem B is denoted ρ^{T_B}, defined by

$$\langle i, j|\rho^{T_B}|k, \ell\rangle = \langle i, \ell|\rho|k, j\rangle.$$

Definition 2.9. A state ρ_{AB} is said to have *positive partial transpose (PPT)* if $\rho^{T_B} \geq 0$.

A very important criterion in quantum information theory that we will widely use is

Peres-Horodecki criterion

The Peres–Horodecki criterion asserts that, given a quantum state ρ:

- If ρ is separable, then it is PPT.
- In dimensions 2×2 and 2×3, the converse is also true: PPT implies separability.

For a proof of the later see the appendix, 6.

Spectral Properties of Quantum Channels

Lemma 2.1. *Theorem 4 in [LG16]. Let $\Phi : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_d(\mathbb{C})$ be a quantum channel. Then:*

- $\sigma(\Phi) \subseteq \{z \in \mathbb{C} : |z| \leq 1\}$
- *The spectrum is closed under complex conjugation:* $\sigma(\Phi^\dagger) = \overline{\sigma(\Phi)}$
- Φ *always admits a fixed point: there exists* $\rho_0 \in \mathcal{D}_d$ *such that* $\Phi(\rho_0) = \rho_0$.
- *The eigenvalues with* $|\lambda| = 1$ *(the* peripheral spectrum*) correspond to isometric dynamics and have trivial Jordan blocks.*

Definition 2.10. The *peripheral spectrum* of Φ, denoted $\sigma_P(\Phi)$, is the set of eigenvalues of Φ with $|\lambda| = 1$, counting multiplicities.

Kadison–Schwarz Inequality

Kadison–Schwarz Inequality
Let $\Phi : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_d(\mathbb{C})$ be a positive unital map. Then for all Hermitian X,

$$\Phi(X)^2 \leq \Phi(X^2).$$

If Φ is completely positive and unital, then

$$\Phi(X^\dagger)\Phi(X) \leq \Phi(X^\dagger X).$$

Wigner's Theorem

Wigner's theorem is a foundational result in quantum mechanics that characterizes transformations preserving transition probabilities. In the context of quantum information theory, it is a key tool for characterizing unitary channels.

Wigner's Theorem
Let $T : \mathcal{H} \to \mathcal{H}$ be a (not necessarily linear) transformation on a Hilbert space $\mathcal{H}$ such that

$$|\langle T(x), T(y)\rangle| = |\langle x, y\rangle| \quad \text{for all } x, y \in \mathcal{H}.$$

Then there exists a real-valued function $\varphi : \mathcal{H} \to \mathbb{R}$ and either a unitary or anti-unitary operator $U : \mathcal{H} \to \mathcal{H}$ such that

$$T(x) = e^{i\varphi(x)}Ux.$$

In finite dimensions, Wigner's theorem implies that any transformation of pure states that preserves transition probabilities must be either unitary or anti-unitary. In the context of quantum channels, this result underlies the equivalence between various characterizations of unitary evolutions (see, e.g., Theorem 3 in [LG16]).

The Bloch Sphere and Qubit Channels

For our experimental tests, see 5, we will need the Blloch representation of qubit states, as well as a representation of qubit channels that follows it. For more details on this see [Gre15].

Lemma 2.2. *Any qubit state $\rho \in \mathcal{D}_2$ can be represented as*

$$\rho = \frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2},$$

where $\vec{n} \in \mathbb{R}^3$, $|\vec{n}| \leq 1$ and $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ is the matrix vector of the three Pauli matrices. The set of such $\vec{n}$ forms the Bloch ball, and pure states lie on its surface.

Lemma 2.3. *Any qubit quantum channel Φ can be represented by a real 4×4 matrix in the Pauli basis:*

$$\Phi(\rho) = \sum_{i,j=0}^{3} \chi_{ij} \sigma_i \rho \sigma_j^\dagger,$$

or more conveniently, in the affine (PTM) form:

$$\Phi\left(\frac{1}{2}(\mathbb{1} + \vec{n} \cdot \vec{\sigma})\right) = \frac{1}{2}(\mathbb{1} + (M\vec{n} + \vec{c}) \cdot \vec{\sigma}),$$

where M is a 3×3 real matrix describing the linear action on the Bloch vector, $\vec{c}$ is a real translation vector and $\vec{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ is the vector of the three Pauli matrices. The full PTM is then given by a 4×4 real matrix:

$$PTM = \begin{pmatrix} 1 & \mathbf{0} \\ \vec{c} & M \end{pmatrix}.$$

3 Entanglement Saving Channels

3.1 Definitions

The natural counterpart to entanglement-breaking channels are those that, in some sense, ”preserve” entanglement under iteration. In this chapter, we introduce and characterize two important classes of quantum channels, following the framework of Lami and Giovannetti [LG16]. These classes are defined by how they affect the entanglement present in a system, and their study reveals connections between the structure of quantum noise and the geometry of quantum channels.

Definition 3.1. A quantum channel $\phi \in \mathbf{CPT}$ is called *entanglement saving* (ES) if $n(\phi) = \infty$, meaning that

$$\phi^n \notin \mathbf{EB} \quad \forall n \in \mathbb{N}.$$

In other words, an ES channel will never break the entanglement of a maximally entangled state when applied locally, no matter how many times it is applied. However, it may still break weaker forms of entanglement, so these channels are not “noiseless” in the strictest sense. ES channels are the complement of eventually entanglement-breaking (EEB) channels within the set of all quantum channels:

$$\mathbf{EEB} \cap \mathbf{ES} = \emptyset, \qquad \mathbf{EEB} \cup \mathbf{ES} = \mathbf{CPT}.$$

This follows from the fact that for any quantum channel ϕ, its entanglement-breaking index $n(\phi)$ is either finite (so $\phi \in \mathbf{EEB}$) or infinite (so $\phi \in \mathbf{ES}$). As we will see, ES channels are extremely rare in the space of all quantum channels.

We are also interested in a more restrictive class, which contains the truly "noiseless" channels:

Definition 3.2. A quantum channel $\phi \in$ **CPT** is called *universal entanglement-preserving* (**UEP**) if its local action inside of a bipartyte system never breaks the entanglement present on it. This means, given any density matrix $\rho_{AB} \in \mathcal{M}_n(\mathbb{C}) \otimes \mathcal{M}_n(\mathbb{C})$, then

$$(\phi \otimes I)\rho_{AB} \notin \text{Sep}(n).$$

Note that this is property is way way stronger than being just **ES**, as an **UEP** channel never breaks the entanglement present in the system regardless of how small it could be.

Lemma 3.1. *Any universal entanglement-preserving channel is, in particular, entanglement saving, i.e.,* $\mathbf{UEP} \subset \mathbf{ES}$.

Proof. Suppose $\phi \in$ **UEP** is not entanglement saving. Then $n(\phi) < \infty$, so there exists an integer $n_0 = n(\phi)$ such that

$$(\phi^{n_0} \otimes I)(|\Omega\rangle\langle\Omega|) \in \text{Sep}(n), \quad \text{but} \quad (\phi^{n_0-1} \otimes I)(|\Omega\rangle\langle\Omega|) \notin \text{Sep}(n),$$

with $|\Omega\rangle$ the maximally entangled state. Setting $\rho_{AB} = (\phi^{n_0-1} \otimes I)(|\Omega\rangle\langle\Omega|)$, we have an entangled state such that $(\phi \otimes I)(\rho_{AB})$ is separable, contradicting the UEP property. □

3.2 Universal Entanglement Preserving Channels

Lami and Giovannetti [LG16] provide a complete characterization of UEP channels:

Theorem (Theorem 17 in [LG16])
The only universal entanglement-preserving (**UEP**) channels are the unitary evolutions.

Recall that a unitary evolution is a channel ϕ such that $\phi(\rho) = U\rho U^\dagger$ for some unitary U.

Before proving this theorem, we need two important results from [LG16]:

Lemma 3.2 (Lemma 16 [LG16]). *Let $Z \in \mathcal{M}_d(\mathbb{C})$ be a non-normal complex matrix (i.e., $[Z, Z^\dagger] \neq 0$). Define*

$$Q(Z) = \begin{pmatrix} I & Z \\ Z^\dagger & Z^\dagger Z \end{pmatrix} \in \mathcal{M}_d(\mathbb{C}) \otimes \mathcal{M}_2(\mathbb{C}).$$

Then $Q(Z)$ is an (unnormalized) entangled state in $\mathcal{M}_d(\mathbb{C}) \otimes \mathcal{M}_2(\mathbb{C})$.

Proof. The proof uses the PPT criterion: the partial transpose of $Q(Z)$ is not positive if $[Z, Z^\dagger] \neq 0$, so $Q(Z)$ is entangled, as PPT is necessary for separability. □

Proposition 3.1 (Proposition 15 [LG16]). *Let ρ_A, ρ_B be density matrices on systems A, B, i.e $\rho_A \in \mathcal{M}_{d_A}(\mathbb{C}), \rho_B \in \mathcal{M}_{d_B}(\mathbb{C})$. Denote by ∂Sep_{AB} the boundary of the set of separable states on the bipartite system AB. Then*

$$\rho_A \otimes \rho_B \in \partial Sep(d_A, d_B) \iff \det \rho_A \cdot \det \rho_B = 0.$$

With this results and some from the preliminaries (see 2), we present a detailed proof for the characterization of **UEP** channels (theorem 3.2).

Proof of 3.2. Note that every unitary evolution is ES, as conjugation by unitaries doesn't affect separability, therefore, we want to proof the other direction of the equivalence.

Let $\phi \in \mathbf{CPT}(n)$ be a UEP channel. We want to show that ϕ must be a unitary evolution. To do so, we divide the proof in 6 different steps.

Step 1: $\phi(\mathbb{1}) > 0$**.**
Suppose that $\phi(\mathbb{1})$ is not strictly positive, then there exists a pure state $|\alpha\rangle$ such that

$$\langle\alpha|\phi(\mathbb{1})|\alpha\rangle = 0.$$

This means that the vector $|\alpha\rangle$ is in the kernel of $\phi(\mathbb{1})$.

Recall that the adjoint channel $\phi^\dagger$ is defined by

$$\mathrm{Tr}[A\phi(B)] = \mathrm{Tr}[\phi^\dagger(A)B]$$

for all matrices A, B. Since ϕ is completely positive and trace-preserving, $\phi^\dagger$ is also completely positive and unital.

Now, observe that

$$\langle\alpha|\phi(\mathbb{1})|\alpha\rangle = \mathrm{Tr}[\phi(\mathbb{1})|\alpha\rangle\langle\alpha|] = \mathrm{Tr}[\mathbb{1}\cdot\phi^\dagger(|\alpha\rangle\langle\alpha|)] = \mathrm{Tr}[\phi^\dagger(|\alpha\rangle\langle\alpha|)].$$

If this trace is zero and $\phi^\dagger$ is positive, it must be that

$$\phi^\dagger(|\alpha\rangle\langle\alpha|) = 0,$$

i.e. $0 \in \sigma(\phi^\dagger)$.

As mentioned in the preliminaries, the spectrum of $\phi^\dagger$ is the complex conjugate of the spectrum of ϕ, i.e., $\sigma(\phi^\dagger) = \overline{\sigma(\phi)}$. Therefore, $0 \in \sigma(\phi)$.

Now, we can take a Hermitian and traceless matrix $X \in \mathcal{M}_n(\mathbb{C})$ such that $\phi(X) = 0$. To see this, take any $Y \in \mathcal{M}_n(\mathbb{C})$ such that $\phi(Y) = 0$, and decompose $Y = A + iB$ with A and B Hermitian. Linearity and Hermiticity preservation (due to positivity) imply that $\phi(A) = \phi(B) = 0$. Also, since ϕ is trace-preserving, $\mathrm{Tr}(A) = \mathrm{Tr}(\phi(A)) = 0$, so A is traceless.

Since X is Hermitian and traceless, it cannot be a multiple of a rank-one projector. Therefore, its support is at least two-dimensional. We can always find another Hermitian matrix Y supported on the same subspace as X such

that $[X, Y] \neq 0$. For example, if X is diagonal in some basis, take Y to have off-diagonal entries in that basis.

Define now, for $\varepsilon > 0$,

$$Z_\varepsilon = X + i\varepsilon Y.$$

Then Z_ε is not normal for $\varepsilon > 0$ (since $[X, Y] \neq 0$), and

$$Q(Z_\varepsilon) = \begin{pmatrix} I & Z_\varepsilon \\ Z_\varepsilon^\dagger & Z_\varepsilon^\dagger Z_\varepsilon \end{pmatrix}$$

is an (unnormalized) entangled state on $\mathcal{M}_d(\mathbb{C}) \otimes \mathcal{M}_2(\mathbb{C})$ by Lemma 3.2.

Applying the channel locally,

$$(I \otimes \phi)(Q(Z_\varepsilon)) = \begin{pmatrix} \phi(\mathbb{1}) & \phi(Z_\varepsilon) \\ \phi(Z_\varepsilon)^\dagger & \phi(Z_\varepsilon^\dagger Z_\varepsilon) \end{pmatrix}.$$

But since $\phi(X) = 0$ and ϕ is linear,

$$\phi(Z_\varepsilon) = \phi(X) + i\varepsilon\phi(Y) = i\varepsilon\phi(Y).$$

Similarly,

$$\phi(Z_\varepsilon^\dagger Z_\varepsilon) = \phi((X - i\varepsilon Y)(X + i\varepsilon Y)) = \phi(X^2 + \varepsilon^2 Y^2 + i\varepsilon[X, Y]).$$

But since ϕ is linear and $\phi(X) = 0$, the dominant term as $\varepsilon \to 0$ is $\phi(X^2)$.

Consider the matrix

$$A_\varepsilon = \begin{pmatrix} 1 & i\varepsilon Y \\ -i\varepsilon Y & Z_\varepsilon^\dagger Z_\varepsilon \end{pmatrix}.$$

Note that $(I \otimes \phi)(Q(Z_\epsilon)) = (I \otimes \phi)(A_\epsilon)$. Then, as $\varepsilon \to 0$,

$$A_\varepsilon \to \begin{pmatrix} 1 & 0 \\ 0 & X^2 \end{pmatrix} = A.$$

Since we are only working inside the support of X, we can project to that subspace and assume that $\det(X) \neq 0$, so $X^2 > 0$. Then, A is a block-diagonal matrix, which is a convex combination of the maximally mixed state and another separable matrix. Because the maximally mixed state is in the interior

of the set of separable states (see Proposition 3.1), then necessarily A must be in the interior also.

Therefore, for sufficiently small $\varepsilon > 0$, A_ε is separable, as the set of separable states is closed. Since the action of $(I \otimes \phi)$ on $Q(Z_\varepsilon)$ produces a separable state for small ε, this contradicts the UEP property, which requires that the output remain entangled for all entangled inputs.

Step 2: Reduction to the Unital Case.
Define the map $\psi(X) = \phi(\mathbb{1})^{-1/2}\phi(X)\phi(\mathbb{1})^{-1/2}$, which is well defined as $\phi(\mathbb{1}) > 0$. Then ψ is unital, completely positive (composition of two **CP** maps), and UEP (conjugation by an invertible matrix doesn't affect separability).

Step 3: Preservation of Non-Invertibility.
Let us first show that if $\rho \geq 0$ and $\det \rho = 0$, then necessarily $\det \psi(\rho) = 0$.

We can assume without loss of generality that ρ is a normalized density matrix. Consider a second system B of dimension $d_B \geq 2$. By Proposition 3.1, the product state $\rho \otimes \frac{1}{d_B}\mathbb{1}_{d_B} \in \mathcal{M}_n(\mathbb{C}) \otimes \mathcal{M}_{d_B}(\mathbb{C})$ lies on the boundary of the set of separable states, that is,

$$\rho \otimes \frac{1}{d_B}\mathbb{1}_{d_B} \in \partial\mathsf{Sep}(n, d_B).$$

This means we can construct a curve of entangled states R_ε (for $0 < \varepsilon \leq 1$) such that

$$\lim_{\varepsilon \to 0^+} R_\varepsilon = \rho \otimes \frac{1}{d_B}\mathbb{1}_{d_B}.$$

Since ψ is UEP, it must be that $(\psi \otimes I)(R_\varepsilon)$ is entangled for all $\varepsilon > 0$. However, by continuity,

$$\lim_{\varepsilon \to 0^+} (\psi \otimes I)(R_\varepsilon) = (\psi \otimes I)\left(\rho \otimes \frac{1}{d_B}\mathbb{1}_{d_B}\right) = \psi(\rho) \otimes \frac{1}{d_B}\mathbb{1}_{d_B}.$$

This state is separable, and in fact, it lies on the boundary of the separable set. By Proposition 3.1, this is only possible if $\det \psi(\rho) = 0$. Thus, ψ preserves non-invertibility for positive matrices.

We now generalize the previous result to all Hermitian matrices. We want to show that for any Hermitian X with $\det X = 0$, it follows that $\det \psi(X) = 0$.

To do this, consider the positive matrix X^2. Clearly, if $\det X = 0$, then $\det(X^2) = 0$. By Step 3, we have $\det \psi(X^2) = 0$, so there exists a vector $|\eta\rangle$ such that

$$\langle\eta|\psi(X^2)|\eta\rangle = 0.$$

Since ψ is positive and unital, we can apply the Kadison–Schwarz inequality (see theorem 2), which gives

$$\langle\eta|\psi(X^2)|\eta\rangle \geq |\langle\eta|\psi(X)|\eta\rangle|^2 .$$

But the left-hand side is zero, so we must have $\langle\eta|\psi(X)|\eta\rangle = 0$. Since $\psi(X)$ is Hermitian, this implies that $\psi(X)$ has a zero eigenvalue, i.e., $\det \psi(X) = 0$.

Step 4: Spectrum Preservation.
Now, we prove preservation of spectrum for Hermitian matrices.

To see why this holds, recall from Step 4 that for any Hermitian X,

$$\det X = 0 \implies \det \psi(X) = 0.$$

Using this, consider an eigenvalue $\lambda \in \sigma(X)$. By definition,

$$\det(X - \lambda\mathbb{1}) = 0.$$

Applying ψ and using the preservation of non-invertibility,

$$\det \psi(X - \lambda\mathbb{1}) = 0.$$

Since ψ is linear and unital,

$$\det(\psi(X) - \lambda\mathbb{1}) = 0,$$

which implies $\lambda \in \sigma(\psi(X))$.

This argument shows that $\sigma(X) \subseteq \sigma(\phi(X))$. To see the other inclusion, we consider Hermitian matrices with non-degenerate spectrum, meaning that it has no repeated eigenvalues, so the spectrum is maximal in size. This implies that $\sigma(X) = \sigma(\phi(X))$ for any Hermitian matrix with non-degenerate spectrum. Now, we can use a result in perturbation theory (see section 6.3 in [HJ85]) which says that the set of Hermitian and non-degenerate matrices

is dense in the set of Hermitian matrices, then, for any Hermitian X we consider a sequence $\{X_\varepsilon\}_{\varepsilon>0}$ of Hermitian and non-degenerate matrices such that $\lim_{\varepsilon\to 0^+} X_\varepsilon = X$.

Since ψ preserves the spectrum exactly for each X_ε,

$$\sigma(\psi(X_\varepsilon)) \equiv \sigma(X_\varepsilon) \quad \forall \varepsilon > 0,$$

and by continuity of eigenvalues with respect to matrix entries,

$$\sigma(\psi(X)) = \lim_{\varepsilon\to 0^+} \sigma(\psi(X_\varepsilon)) = \lim_{\varepsilon\to 0^+} \sigma(X_\varepsilon) = \sigma(X).$$

This preservation will allow us to access the conditions of Wigner's theorem.

Step 5: Preservation of purity and inner product of pure states
We now show that ψ sends pure states to pure states, and moreover, it preserves the modulus of the inner product between any two pure states. This step is crucial to prepare the ground for the application of Wigner's theorem.

First, for any pure state $|\alpha\rangle$, the corresponding rank-one projector satisfies $\sigma(|\alpha\rangle\langle\alpha|) = \{1, 0, \ldots, 0\}$. Since ψ preserves the spectrum of Hermitian matrices (as shown in the previous step), we have

$$\sigma(\psi(|\alpha\rangle\langle\alpha|)) = \{1, 0, \ldots, 0\}$$

which implies that $\psi(|\alpha\rangle\langle\alpha|)$ is also a rank-one projector, i.e., a pure state. Thus, for every pure state $|\alpha\rangle$, there exists a pure state $|\alpha'\rangle$ such that

$$\psi(|\alpha\rangle\langle\alpha|) = |\alpha'\rangle\langle\alpha'|.$$

Next, consider two pure states $|\alpha\rangle$ and $|\beta\rangle$, and let $|\alpha'\rangle$, $|\beta'\rangle$ be their images under ψ. The spectrum of the sum of two rank-one projectors is

$$\sigma(|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta|) = \{1 + |\langle\alpha|\beta\rangle|, 1 - |\langle\alpha|\beta\rangle|, 0, \ldots, 0\}.$$

Since ψ preserves the spectrum,

$$\sigma(\psi(|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta|)) = \sigma(|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta|).$$

But by linearity and preservation of purity

$$\psi(|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta|) = |\alpha'\rangle\langle\alpha'| + |\beta'\rangle\langle\beta'|,$$

so

$$\sigma(\psi(|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta|)) = \{1 + |\langle\alpha'|\beta'\rangle|, 1 - |\langle\alpha'|\beta'\rangle|, 0, \ldots, 0\}.$$

Comparing the spectra, we deduce

$$|\langle\alpha|\beta\rangle| = |\langle\alpha'|\beta'\rangle|.$$

Therefore, ψ maps pure states to pure states and preserves the modulus of their inner products. This property is exactly the hypothesis of Wigner's theorem (see theorem 2), which will allow us to conclude that ψ is either a unitary or anti-unitary transformation (the anti-unitary case is excluded by complete positivity in the next step).

Step 6: Application of Wigner's Theorem.
Thanks to the preservation of the modulus of inner products between pure states (as shown in the previous step), the hypotheses of Wigner's Theorem are satisfied, so we can claim that for every vector $|\alpha\rangle$, the image under ψ is of the form

$$|\alpha'\rangle = e^{i\varphi(\alpha)}U|\alpha\rangle \quad \text{or} \quad |\alpha'\rangle = e^{i\varphi(\alpha)}U|\alpha^*\rangle,$$

where U is a unitary operator and $|\alpha^*\rangle$ denotes the complex conjugate of $|\alpha\rangle$ in some basis (note that an anti-unitary can be expressed as the composition of a unitary and complex conjugation).

Therefore, for any pure state,

$$\psi(|\alpha\rangle\langle\alpha|) = U|\alpha\rangle\langle\alpha|U^\dagger \quad \text{or} \quad \psi(|\alpha\rangle\langle\alpha|) = U|\alpha^*\rangle\langle\alpha^*|U^\dagger = U|\alpha\rangle\langle\alpha|^T U^\dagger.$$

This extends by linearity (one-dimensional projections span the full matrix space) to all input matrices X:

$$\psi(X) = UXU^\dagger \quad \text{or} \quad \psi(X) = UX^TU^\dagger.$$

However, the second option (involving the transpose) must be discarded, because the matrix transposition is only a positive map, not completely positive,

and ψ is completely positive by construction. Thus, only the unitary case remains:

$$\psi(X) = UXU^\dagger.$$

Recall that ψ was defined as

$$\psi(X) = \phi(\mathbb{1})^{-1/2}\phi(X)\phi(\mathbb{1})^{-1/2}.$$

Substituting back, we find

$$\phi(X) = \phi(\mathbb{1})^{1/2}UXU^\dagger\phi(\mathbb{1})^{1/2}.$$

Since ϕ is trace-preserving, it must be that $U^\dagger\phi(\mathbb{1})U = \mathbb{1}$, so $\phi(\mathbb{1}) = \mathbb{1}$. Therefore,

$$\phi(X) = UXU^\dagger$$

for all input matrices X, as claimed. □

This characterization of universal entanglement-preserving channels as unitary evolutions confirms a long-held physical intuition: only noiseless, reversible dynamics can preserve all forms of entanglement indefinitely. While this was assumed in quantum information theory, Lami and Giovannetti's required all this machinery in order to prove it rigorously for first time.

3.3 Entanglement Saving Channels

We now turn to entanglement-saving (ES) channels, this broader class that preserves entanglement under iteration but may break weaker forms of quantum correlations. Lami and Giovanetti also present a big theorem regarding the characterization of this channels, which we were really interested in. They specifically provide an almost complete characterization under the condition $\det\phi \neq 0$. Before presenting its proof, we require two supporting results from the same paper:

Proposition 3.2 (Corollary 14 in [LG16]). *Let $\phi \in \mathbf{CPt}_d$ be a quantum channel satisfying $|\sigma_P(\phi)| \geq 2$ (see thefinition 2.10). Then there exists an integer $1 \leq n \leq d$ such that 1 belongs to $\sigma_P(\phi^n)$ with multiplicity strictly greater than 1.*

Proposition 3.3 (Corollary 19 in [LG16]). *Let $\phi \in \mathbf{EB}_d$ be an entanglement-breaking channel with a semipositive fixed point, meaning that there exist a semidefinite positive matrix M in the input space of ϕ such that $\phi(M) = M$. Then $\dim \ker \phi \geq 2(d-1)$ and, in particular, $\det \phi = 0$.*

Lemma 3.3 (Lemma 20 in [LG16]). *Let $\phi \in \mathbf{PT}$ be a positive trace-preserving map with multiplicity of eigenvalue 1 strictly greater than 1. Then ϕ admits a semipositive fixed point.*

We now state and prove the main classification theorem for ES channels.

Theorem (Theorem 21 in [LG16])
Let $\phi \in \mathbf{CPT}_d$ satisfy $\det \phi \neq 0$. Then the following are equivalent:

1. ϕ is entanglement-saving (**ES**).
2. ϕ has a semipositive fixed point or $|\sigma_P(\phi)| \geq 2$.
3. Some ϕ^n $(1 \leq n \leq d)$ has a semipositive fixed point

Proof. We prove the implications cyclically:

Proof of (1) $\Rightarrow$ (2)
We prove that if ϕ is entanglement-saving (ES), then it must have either a semipositive fixed point or at least two peripheral eigenvalues ($|\sigma_P(\phi)| \geq 2$). The argument holds independently of the determinant condition and proceeds by contradiction.

Suppose ϕ has a *strictly positive* fixed point $\rho_0 > 0$ and its peripheral spectrum is trivial: $\sigma_P(\phi) = \{1\}$ (i.e., no eigenvalues of modulus 1 except 1 itself, with multiplicity 1).

By spectral properties from lemma 2.1 in the preliminaries, all other eigenvalues satisfy $|\lambda| < 1$. Thus, as $n \to \infty$, ϕ^n converges to the completely depolarizing channel, i.e

$$\phi^n \to D_{\rho_0}, \quad \text{where} \quad D_{\rho_0}(X) = \rho_0 \operatorname{Tr} X.$$

This convergence occurs because:

- The spectral radius of non-peripheral eigenvalues is < 1, so their contributions vanish under iteration.
- The unique fixed point ρ_0 dominates the asymptotic behavior, "depolarizing" all inputs to ρ_0.

The Choi-Jamiołkowski isomorphism is linear and continuous. Thus,

$$\lim_{n\to\infty} R_{\phi^n} = R_{D_{\rho_0}} = \rho_0 \otimes \frac{\mathbb{1}_d}{d}.$$

Since $\rho_0 > 0$, Proposition 3.1 implies that $\rho_0 \otimes \frac{\mathbb{1}_d}{d}$ lies in the *interior* of the separable set $\operatorname{Sep}(d)$. Consequently, for sufficiently large n, R_{ϕ^n} is separable. This means ϕ is **EEB**, contradicting the ES property.

Proof of (2) $\Rightarrow$ (3)
First, if ϕ admits a semipositive fixed point ρ_0 ($\det \rho_0 = 0$), then (3) holds trivially for $n = 1$, as $\phi^1 = \phi$ fixes ρ_0.

On the other hand, suppose ϕ has no semipositive fixed point but satisfies $|\sigma_P(\phi)| \geq 2$. By proposition 3.2, there exists an integer $1 \leq n \leq d$ such that ϕ^n has eigenvalue 1 with multiplicity at least 2. Since ϕ^n is completely positive and trace-preserving, Lemma 3.3 implies that ϕ^n admits a semipositive fixed point.

Proof of (3) $\Rightarrow$ (1)
This implication is the one that really uses the assumption $\det \phi \neq 0$.

Suppose ϕ is *not* ES. Then by definition, there exists some finite $N \in \mathbb{N}$ such that ϕ^N is entanglement-breaking (EB), hence ϕ^K is also EB for any $K \geq N$.

Now, by hypothesis (3), ϕ^{n_0} has a semipositive fixed point ρ_0 for some $n_0 < d$. Since ρ_0 is fixed by ϕ^{n_0}, it is also fixed by all powers of ϕ^{n_0}:

$$\phi^{kn_0}(\rho_0) = (\phi^{n_0})^k(\rho_0) = \rho_0.$$

Then, we conclude that there exist an integer N_0 such that ϕ^{N_0} is EB and has a semipositive fixed point. Therefor, by proposition 3.3 we get that $\det(\phi^{N_0}) = 0$, which implies that $\det(\phi) = 0$, contradicting our assumption. □

This theorem provides a complete characterization of entanglement-saving (ES) channels under the condition $\det \phi \neq 0$. A consequence of this characterization is that ES channels form a set of measure zero within the space of all quantum channels. This result follows from algebraic geometry arguments. The conditions for ES behavior (existence of a semipositive fixed point or $|\sigma_P(\phi)| \geq 2$) define algebraic varieties of lower dimension in the space of quantum channels. To see this, we refer to the following result in [HMR21]:

Proposition 3.4. *Let $\mathcal{M} \subset \mathbb{R}^m$ be an algebraic manifold of dimension d, and let $\mathcal{N}$ be an algebraic sub-variety of $\mathcal{M}$ of dimension at most $d-1$. Let μ be a measure on $\mathcal{M}$ locally equivalent to the Lebesgue measure. Then $\mu(\mathcal{N}) = 0$.*

Then, one can see that all conditions, namely the left out channels with $\det(\phi) = 0$, channels with a semipositive fixed point and finally the ones with more than two peripheral points in the spectrum, define algebraic sub-varieties of dimension strictly less than $\dim(\mathbf{CPT}_n)$, the dimension of the set of quantum channels.

As we struggled to understand the theory behind this measure-zero argument, we performed some experimental tests, that will be explained in detail in the chapter 5. We wanted to see that, as a consequence of theorem 3.3, a random quantum channel will be eventually entanglement breaking with probability 1, due to its complement (ES channels) having measure zero.

When doing so, we decided to stick to the case of qubit channels. With this we refer to quantum channels in dimension 2, namely $\phi : \mathcal{M}_2(\mathbb{C}) \to \mathcal{M}_2(\mathbb{C})$. In this dimension, we know that the PPT criteria is sufficient for separability (see

theorem 2) and that the characterization of entanglement saving channels is complete (see next section).

3.3.1 Qubit channels

This last statement is also proved in Lami's and Giovanetti's paper, mainly using this result:

Lemma 3.4 (Lemma 22 in [LG16]). *For any qubit channel $\phi \in \mathbf{CPT}_2$, $\det \phi = 0$ implies ϕ is entanglement-breaking.*

This lemma eliminates the need for the $\det \phi \neq 0$ assumption in Theorem 3.3 for qubits. Combined with the explicit characterization, Lami and Giovanetti prove the following meaningful result:

Theorem (Theorem 24 in [LG16])
A qubit channel $\phi \in \mathbf{CPT}_2$ is entanglement-saving if and only if ϕ fixes a pure state or inverts a pure state in the Bloch sphere.

In the theorem, inversion of pure states refers to its geometric geometric interpretation in the bloch sphere. Given a qubit state, so a density matrix $\rho \in \mathcal{M}_2(\mathbb{C})$, using its representation in the Bloch sphere (see lemma 2.2), $\rho = \frac{\mathbb{1}+n\cdot\vec{\sigma}}{2}$, we say that ϕ inverts ρ if

$$\phi(\rho) = \frac{\mathbb{1} - n \cdot \vec{\sigma}}{2}$$

In the chapter 5, we will explain how we tried to implement a function that checks whether a quantum channel is entanglement saving and also how we ran some tests on properties of random quantum channels, basically checking whether they are or not EEB.

4 The PPT Squared Conjecture

PPT Squared Conjecture - Version 1
If a linear map $T : M_d \to M_d$ is both completely positive and completely copositive, then its square $T \circ T$ is entanglement breaking.

This conjecture was originally stated by M. Christandl at the Banff International Research Station workshop in 2012. It has since attracted a lot of attention in the Quantum Information community. While it is generally thought that the conjecture will eventually be disproven, many years after its publication it still stands. Moreover, the list of examples where the PPT squared conjecture holds has reached a notable amount. In the following section, we will try to dissect this beautiful conjecture, which strikes by its simplicity and follow the question why just two applications of any copositive map should always lead to a separable state.

4.1 The importance of PPT

In this short section, we will provide a brief overview of the importance of positive partial transpose in the context of the separability problem. In 2001, Horodecki et al. [HHH96]
proved the following equivalent definition for a state to be separable.

Theorem (M. Horodecki, P. Horodecki and R. Horodecki, 1996)
Let ρ act on the Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$. Then ρ is separable if and only if for any positive map $\Lambda : B(\mathcal{H}_2) \to B(\mathcal{H}_1)$, the operator $(I \otimes \Lambda)\rho$ is positive.

While it is not representative of the chronological order of the results, the above statement can be seen as generalisation of the theorem below.

Theorem (Peres, 1996)
Let ρ be separable then the partial transpose $\rho^{T_i} \geq 0$ for $i = A, B$.

While the Peres criterion, often also refered to as the PPT criterion, only gives a necessary condition for separability for bipartite systems of arbitrary dimension, it has yet proven to be a powerful tool in the quest to determine if an arbitrary state is separable or not. It's easy to check if a density matrix is PPT and at the same time the Peres criterion is stronger than for example the reduction criterion [HH97]. Note that for system of size 2×2 and 2×3, it's also sufficient. The proof for this result can be found in the appendix as we wanted to look deeper into this result (6). The notion of the PPT criterion and separability are clearly connected. The conjecture of M. Christandl which we will discuss in detail in this chapter, takes this connection one step further.

4.2 Equivalences of the PPT squared conjecture

In the paper [CMW19] three equivalent conjectures are given which allow to draw connections to different fields in quantum information theory. They will also allow us to give us a physical interpretation of the conjecture and its implications later on.

Composition of different linear maps (Version 2)
The first equivalent formulation may be seen as a generalization of version 1. Note that in the original conjecture entails the application of linear map with equal input and output dimension, twice. We will now show that this implies a more general statement, specifically that the composition of any two completely positive and completely copositive maps is entanglement breaking.

PPT Squared Conjecture - Version 2
For any pair of linear maps $T_1 : M_{d_1} \to M_{d_2}$ and $T_2 : M_{d_2} \to M_{d_3}$ that are both completely positive and completely copositive, the composition $T_2 \circ T_1$ is entanglement breaking.

Proof of equivalence of version 1 and 2: First notice that the *only if* direction is obvious. To show that the other direction also holds, we will perform a proof by contraposition. Assume that there exists a pair of linear maps $T_1 : M_{d_1} \to M_{d_2}$ and $T_2 : M_{d_2} \to M_{d_3}$ that both completely positive and completely copositive, but their composition is *not* entanglement breaking. There two key concepts to this proof. The first is the isometric embedding of $\mathbb{C}^{d_i} \to \mathbb{C}^d$ for $i = 1, 2$ with $d = \max(d_1, d_2, d_3)$. The corresponding isometries are denoted by the maps $V_1 : \mathbb{C}^{d_1} \to \mathbb{C}^d$ and $V_2 : \mathbb{C}^{d_2} \to \mathbb{C}^d$. These allow us to define linear maps $\tilde{T}_1 : M_d \to M_d$ and $\tilde{T}_2 : M_d \to M_d$ with equal input and output dimensions.

$$\tilde{T}_1(X) = T_1(V_1^\dagger X V_1) \oplus 0_{(d-d_2)}$$
$$\tilde{T}_2(X) = T_2(V_2^\dagger X V_2) \oplus 0_{(d-d_3)}$$

for any input matrix $X \in M_d$. The action of these maps is a projection onto the smaller subspace, the application of the original maps T_1, T_2 and finally a trivial lifiting. It's important to point out that both maps are still completely positive and completely copositive. Each of the two newly defined maps correspond to a composition of three completely positive maps and is therefore again completely positive. The map $\psi : M_d \to M_{d_i}$ defined by $\psi(X) = V_i^\dagger X V_i$ is already in its Kraus decomposition (with $K = V_i^\dagger$). By definition T_1 is completely positive and trivially the trivial lifting is too as it corresponds to the same physical process in a greater Hilbert space. As the composition with a single completely copositive map is again completely copositive and T_1, T_2 are completely copositive by definition, then so are $\tilde{T}_1$, $\tilde{T}_2$.
The second key concept is the introduction of a so-called switch map $T : M_d \otimes M_2 \to M_d \otimes M_2$ defined as

$$\begin{aligned} T(X) = \tilde{T}_1\Big((\mathbb{I}_d \otimes \langle 1|)X(\mathbb{I}_d \otimes |1\rangle)\Big) \otimes |2\rangle\langle 2| \\ + \tilde{T}_2\Big((\mathbb{I}_d \otimes \langle 2|)X(\mathbb{I}_d \otimes |2\rangle)\Big) \otimes |1\rangle\langle 1| \end{aligned}$$

for any $X \in M_d \otimes M_2$ with $\mathbb{I}_d$ denoting the identity in M_d. The added two dimensional ancilla system acts as a short term memory, preventing $\tilde{T}_1$ (or $\tilde{T}_2$)

to be applied twice in a row if T is applied multiple times. In other words, if $X = Y \otimes |1\rangle\langle 1| \in M_d \otimes M_2$ then

$$T(X) = \tilde{T}_1(Y) \otimes |2\rangle\langle 2|$$

as $|1\rangle, |2\rangle$ are two orthogonal states in the coputational basis. With this result, it is now easy to see that applying the channel T twice yields

$$\begin{aligned} T \circ T(X) = \tilde{T}_1 \circ \tilde{T}_2 \left((\mathbb{I}_d \otimes \langle 2|) X (\mathbb{I}_d \otimes |2\rangle) \right) \otimes |2\rangle\langle 2| \\ + \tilde{T}_2 \circ \tilde{T}_1 \left((\mathbb{I}_d \otimes \langle 1|) X (\mathbb{I}_d \otimes |1\rangle) \right) \otimes |1\rangle\langle 1| \end{aligned}$$

Note again that T is still completely positive as it corresponds to a composition of $\tilde{T}_i$ and a measurement and preparation channel on the ancilla system. The complete copositivity follows from the same argument as before. Finally, note that

$$\begin{aligned} T \circ T(V_1 Y V_1^\dagger \otimes |1\rangle\langle 1|) &= \tilde{T}_2 \circ \tilde{T}_1(V_1 Y V_1^\dagger) \otimes |1\rangle\langle 1| \\ &= \tilde{T}_2(T_1(V_1^\dagger V_1 Y V_1^\dagger V_1) \oplus 0_{(d-d_2)}) \otimes |1\rangle\langle 1| \\ &= \tilde{T}_2(T_1(Y)) \oplus 0_{(d-d_2)}) \otimes |1\rangle\langle 1| \\ &= T_2(V_2^\dagger (T_1(Y) \oplus 0_{(d-d_2)}) V_2) \oplus 0_{(d-d_3)} \otimes |1\rangle\langle 1| \\ &= T_2 \circ T_1(Y) \oplus 0_{(d-d_3)} \otimes |1\rangle\langle 1| \end{aligned}$$

for any $Y \in M_{d_1}$. By our assumption, the channel $T_2 \circ T_1$ is not entanglement breaking and therefore neither is $T \circ T$. □

Connection to local entanglement annihilation (Version 3)

The second reformulation of the conjecture offers a connection to 2-locally entanglement annihilating channels. A linear map $T : \mathcal{M}_{d_1} \to \mathcal{M}_{d_2}$ is called *2-locally entanglement annihilating* (2-LEA) if the image $(T \otimes T)(X)$ is separable for any positive matrix $X \in M_{d_1} \otimes M_{d_1}$. Clearly, entanglement breaking maps are also 2-locally entanglement annihilating maps, but the other implication is not necessarily true [FRZ12].

PPT Squared Conjecture - Version 3
For any pair of linear maps $T_1 : M_{d_1} \to M_{d_2}$ and $T_2 : M_{d_3} \to M_{d_4}$ that are completely positive and completely copositive, the image $(T_1 \otimes T_2)(X)$ is separable for any positive matrix $X \in (M_{d_1} \otimes M_{d_3})^+$.

Proof of equivalence of version 2 and 3: We will first proof the *if* direction and assume that version 2 holds. The set of states is convex and any state can be written as a convex combination of pure states. Hence, it suffices to show that $(T_1 \otimes T_2)(|\psi\rangle\langle\psi|)$ is separable for any pure state $|\psi\rangle \in \mathbb{C}^{d_1} \otimes \mathbb{C}^{d_3}$. By the first trick in 2 for the maximally entangled state we can write $|\psi\rangle = (\mathbb{I}_{d_1} \otimes A)|\Omega_{d_1}\rangle$ with a linear transformation $A : \mathbb{C}^{d_1} \to \mathbb{C}^{d_3}$. Together with the second trick in 2, we arrive to

$$(T_1 \otimes T_2)(|\psi\rangle\langle\psi|) = \Big(T_1 \otimes (T_2 \circ \mathrm{Ad}_A)\Big)(|\Omega_{d_1}\rangle\langle\Omega_{d_1}|) \tag{4.1}$$
$$= \Big(\mathrm{id}_{d_1} \otimes (T_2 \circ \mathrm{Ad}_A)\Big)\Big(T_1 \otimes \mathrm{id}_{d_1}\Big)(|\Omega_{d_1}\rangle\langle\Omega_{d_1}|) \tag{4.2}$$
$$= [\mathrm{id}_{d_2} \otimes (T_2 \circ \mathrm{Ad}_A \circ \vartheta_{d_1} \circ T_1^* \circ \vartheta_{d_2})]\,(|\Omega_{d_2}\rangle\langle\Omega_{d_2}|) \tag{4.3}$$

Note that the map $\mathrm{Ad}_A \circ \vartheta_{d_1} \circ T_1^* \circ \vartheta_{d_2} : M_{d_2} \to M_{d_3}$ is still compeltely positive. The adjoint T^* of a completely positive map T is still completely positive as well as the map $\vartheta \circ T \circ \vartheta$. Furthermore, remember that the composition of two maps is again completely copositive if at least one of the maps is completely copositive. Hence, $\mathrm{Ad}_A \circ \vartheta_{d_1} \circ T_1^* \circ \vartheta_{d_2} : M_{d_2} \to M_{d_3}$ is again completely copositive. By our assumption the map $T_2 \circ \mathrm{Ad}_A \circ \vartheta_{d_1} \circ T_1^* \circ \vartheta_{d_2}$ is entanglement breaking and $(T_1 \otimes T_2)(|\psi\rangle\langle\psi|)$ is separable.
To prove the *only if* direction, assume that version 3 is true. We then choose a completely positive and completely copositive map $T : M_d \to M_d$ and define the maps $T_{1,2} : M_d \to M_d$ by $T_1(X) = (\vartheta_d \circ T^* \circ \vartheta_d)(X)$ and $T_2(X) = T(X)$ for any $X \in M_d$. Notice that both T_1 and T_2 are completely positive and completely copositive. Similarly to before, we obtain

$$(T_1 \otimes T_2)(\omega_d) = \Big((\vartheta_d \circ T^* \circ \vartheta_d) \otimes T\Big)(\omega_d) = (\mathrm{id}_d \otimes T^2)(\omega_d) \tag{4.4}$$

Notice that the right side is the Choi matrix of the map T^2. By assumption $(T_1 \otimes T_2)(\omega_d)$ is separable and so is the Choi matrix. As map is entanglement breaking if and only if its Choi is separable, we conclude that T^2 is entanglement breaking. □

Decomposability of certain positive maps (Version 4)
The last reformulation is particularly useful as it connects the PPT squared conjecture to the decomposibilty of positive maps and allows the search for a counter example using semidefinite programming.

Definition (Decomposibilty)
A positive map $P : M_{d_1} \to M_{d_2}$ is called *decomposable* if $P = T_1 + \vartheta_{d_2} \circ T_2$, for some CP maps $T_1, T_2 : M_{d_1} \to M_{d_2}$.

PPT Squared Conjecture - Version 4
For any completely positive and completely copositive map $T : M_{d_1} \to M_{d_2}$ and any positive map $P : M_{d_2} \to M_{d_3}$, the composition $P \circ T : M_{d_1} \to M_{d_3}$ is decomposable.

To prove the final equivalence, we will need the two Lemmas below. The first is an equivalent definition by Størmer of what it means for a map to be decomposable. The second is simply a reformulation of theorem 4.1 in terms of entanglement breaking maps.

Lemma (Størmer [Sto82])
A linear map $T : M_{d_1} \to M_{d_2}$ is *decomposable* if and only if $(\mathrm{id}_{d_A} \otimes T)(X) \geq 0 \quad$ for any $X \in (M_{d_A} \otimes M_{d_B})^+$ with $X^\Gamma \geq 0$ where Γ denotes the partial transpose.

Lemma (Horodecki et al. [HHH96])
A linear map $T : M_{d_1} \to M_{d_2}$ is *entanglement breaking* if and only if for any positive map $P : M_{d_2} \to M_{d_1}$, the composition $P \circ T : M_{d_1} \to M_{d_1}$ is completely positive.

Proof of equivalence of version 2 and 4: We will start with the *if* direction and assume that version 2 of the conjecture holds. Consider a map $T : M_{d_1} \to M_{d_2}$ that is completely positive and completely copositive, and a map $P : M_{d_2} \to M_{d_3}$ that is positive. For any completely positive and completely copositive map $S : M_{d_3} \to M_{d_1}$, the composition $T \circ S : M_{d_3} \to M_{d_2}$

is entanglement breaking by assumption. By Lemma 4.2, the composition $P \circ T \circ S$ is completely positive. Since this holds for any completely positive and completely copositive map S, Lemma 4.2 shows that $P \circ T$ must be decomposable.

We will prove the *only if* direction via contraposition. Assume the maps $T : M_{d_1} \to M_{d_2}$ and $S : M_{d_3} \to M_{d_1}$ are completely positive and completely copositive, but $T \circ S : M_{d_3} \to M_{d_2}$ is *not* entanglement. By Lemma 4.2, there exists a positive map $P : M_{d_2} \to M_{d_3}$ such that $P \circ T \circ S$ is *not* completely positive. By Lemma 4.2, $P \circ T$ *can't* be decomposable. □

4.3 Example for the PPT squared conjecture

Finally, we want to give an example of maps that are completely positive and completely copositive for whicht the PPT squared conjecture holds. Note that if such a map is also entanglement breaking the conjecture trivially holds. The interesting cases are the ones where T is not entanglement breaking, but $T \circ T$ is. In our case, we consider completely positive maps $T : \mathcal{M}_d \otimes \mathcal{M}_d \to \mathcal{M}_d \otimes \mathcal{M}_d$ such that

$$(\mathrm{Ad}_U \otimes \mathrm{Ad}_V) \circ T \circ (\mathrm{Ad}_{U^T} \otimes \mathrm{Ad}_{V^\dagger}) = T$$

for all unitaries $U, V \in \mathcal{U}_d$. Completely positive maps with such a symmetry correspond to Choi matrices with the following symmetry

$$(U \otimes U \otimes V \otimes \bar{V})\, C_T\, (U \otimes U \otimes V \otimes \bar{V})^\dagger = C_T$$

To see this, note that the Choi matrix for such maps is

$$C_T = (\mathrm{id}_{d^2} \otimes T)\, |\Omega\rangle\langle\Omega| = (\vartheta_{d^2} \circ (\mathrm{Ad}_{U^T} \otimes \mathrm{Ad}_{V^\dagger})^* \circ \vartheta_{d^2}) \otimes (\mathrm{Ad}_U \otimes \mathrm{Ad}_V)\, |\Omega\rangle\langle\Omega|$$

where we have used in the Lemma that we used in the proof of the second equivalence. Further note that $\mathrm{Ad}_U^* = \mathrm{Ad}_{U^\dagger}$ and $(\vartheta_{d^1} \circ \mathrm{Ad}_U^* \circ \vartheta_{d^1})(X) = (U^\dagger X^T U)^T = \mathrm{Ad}_{U^T}(X)$ with $X \in M_d$. Putting this all together, we arrive to

$$C_T = (\mathrm{Ad}_U \otimes \mathrm{Ad}_{V^\dagger} \otimes \mathrm{Ad}_U \otimes \mathrm{Ad}_V)\, |\Omega\rangle\langle\Omega|$$

The last thing that remains to be done is a reordering of the tensor factors ($\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3 \otimes \mathcal{H}_4 \to \mathcal{H}_1 \otimes \mathcal{H}_3 \otimes \mathcal{H}_4 \otimes \mathcal{H}_2$). Both spaces are isomorphic to each other. The set of states with this specific symmetry is known to contain entangled states with positive partial transpose [VW02]. Yet the corresponding completely positive and completely copositive maps to any pair of states from this set satisfies the PPT squared conjecture. In other words, for any pair of completely positive and completely copositive maps $T_1, T_2 : \mathcal{M}_d \otimes \mathcal{M}_d \to \mathcal{M}_d \otimes \mathcal{M}_d$ that satisfy 4.3 their composition $T_2 \circ T_1$ is entanglement breaking.

Proof: Note that for any unitaries $U, V \in \mathcal{U}_d$

$$\begin{aligned} T_2 \circ T_1 &= (\mathrm{Ad}_U \otimes \mathrm{Ad}_V) \circ T_2 \circ (\mathrm{Ad}_{U^T} \otimes \mathrm{Ad}_{V^\dagger}) \circ (\mathrm{Ad}_{\bar{U}} \otimes \mathrm{Ad}_V) \circ T_1 \circ (\mathrm{Ad}_{U^\dagger} \otimes \mathrm{Ad}_{V^\dagger}) \\ &= (\mathrm{Ad}_U \otimes \mathrm{Ad}_V) \circ T_2 \circ T_1 \circ (\mathrm{Ad}_{U^\dagger} \otimes \mathrm{Ad}_{V^\dagger}) \end{aligned}$$

as $(\mathrm{Ad}_{U^T} \otimes \mathrm{Ad}_{V^\dagger}) \circ (\mathrm{Ad}_{\bar{U}} \otimes \mathrm{Ad}_V) = \mathrm{id}_{d^2}$. Analogously to above, a map with such a symmetry corresponds to

$$(U \otimes \bar{U} \otimes V \otimes \bar{V})\, C_{T_2 \circ T_1}\, (U \otimes \bar{U} \otimes V \otimes \bar{V})^\dagger = C_{T_2 \circ T_1}$$

At this point, we would like to point out that the authors of the paper arrive to a different result namely

$$(U \otimes U \otimes V \otimes V)\, C_{T_2 \circ T_1}\, (U \otimes U \otimes V \otimes V)^\dagger = C_{T_2 \circ T_1}$$

This is crucial for the proof as it relies on a result from [VW02] showing that positive matrices with a symmetry as above 4.3 is separable if and only if they have positive partial transpose. This would show that $T_2 \circ T_1$ is entanglement breaking as the Choi matrix of a completely copositive map is positive under partial transposition. We claim that it's not possible to find a Choi matrix with the needed symmetry for any possible symmetry of T_1 and T_2. In this case, the result reduces to real valued unitaries $U, V \in \mathcal{U}_d$ and at the current moment we have not investigated yet if there are entangled states with positive partial transpose in this subset of states too. If this wasn't the case, then in this form the example wouldn't be one.

4.4 Physical implications of the conjecture

If the PPT squared conjecture was indeed true, this would pose limitations for quantum key distribution (QKD) protocols or specifically quantum key repeaters. Many of such protocols rely on shared entanglement between two parties in form of an EPR pair (the maximally entangled state in a qubit system). As the distribution of such entanglement over long distances is highly challenging and limited due to a constant influence of the environment, one uses a so-called quantum repeater. This quantum repeater is operated by a third party, Charlie, a telecom provider that is situated between Alice and Bob who want to execute a QKD protocol. Such a device must perform entanglement distillation: a protocol that allows to distill an EPR pair from an entangled mixed state. Remember that most states in the lab are mixed due to interaction with the environment. This EPR pair is then used to execute a quantum teleportation protocol. However, there are entangled mixed states which can't be distilled, so-called bound states [HHH98]. This set of states includes entangled states with positive partial transpose. In [Bäu+15] the authors follow the question how useful such states can be for QKD. If the PPT squared conjecture was true, then this would imply the following

Assume that Alice and Charlie share a PPT state, and that Bob and Charlie share a PPT state. Then the state of Alice and Bob, conditioned on any measurement by Charlie, is always separable.

In other words, entangled states with positive partial transpose are not useful for quantum key repeaters. To see that the above statement is true, notice that the system can be described by two completely copositive maps T_A, T_B acting on Alice's and Bob's subsystem

$$\rho'_{A,B,C} = (T_A \otimes T_B \otimes \mathrm{id}_C)\rho_{A,B,C}$$

The reduced density matrix of Alice and Charlie is then equal to $\rho'_{A,C} = \mathrm{Tr}_B(\rho'_{A,B,C})$ which is a PPT state (analogously for Bob and Charlie). The third version of the PPT squared conjecture then tells us that $\rho'_{A,B,C}$ will be separable with respect to the partition A|BC (or AC|B). Hence, conditioned on any measurement by Charlie the state of Alice and Bob is always separable.

5 Numerical Experiments

In this chapter, we present numerical investigations into the behavior of random quantum channels under composition, with a particular focus on their tendency to become entanglement-breaking (EB) after repeated application. The goal is to test and support the theoretical understanding of *eventually entanglement-breaking* (EEB) and *entanglement saving* (ES) channels using the intuition from the papers we have read.

5.1 Eventually Entanglement Breaking

5.1.1 Methodology and Tools

We employ the `Qiskit` framework to generate and manipulate random quantum channels and simulate their repeated application. In particular, we use Qiskit's `random_quantum_channel` function to sample random completely positive and trace-preserving (CPT) maps. This function draws channels from an ensemble consistent with the Haar measure over Kraus operators, yielding a diverse and representative sample of physically meaningful noise models.

To determine whether a given channel is EB, we use the following strategy: since all EB channels are positive under partial transpose (PPT) when acting on one side of a maximally entangled state, we use the partial transpose criterion on the associated Choi matrix. The following utility functions, implemented in Python, allow us to perform this check efficiently:

```
def is_ppt(choi):
    """Check if the Choi matrix is PPT (Positive Partial
       Transpose)"""
    dm = DensityMatrix(choi.data)
    dm_pt = dm.partial_transpose(qargs=[1])
    return all(np.linalg.eigvalsh(dm_pt.data) >= 0)
```

```
def is_eppt(choi, app):
    """Check if the Choi matrix is EEB (Eventually Entanglement
        Breaking)"""
    for _ in range(app):
        choi = choi.compose(choi)
    return is_ppt(choi)
```

Listing 5.1: Check if a Choi matrix is PPT or Eventually PPT

The aim of this experiments is to focus only on qubit channels, so that we can use the PPT criteria (2). Therefore, to see that a qubit channel is entanglement breaking (that is in general computationally hard) it's enough to check whether it is PPT. In order to do so, we have developed two functions, one that checks if a qubit channel is PPT and one that checks if it is what we call Eventually PPT (EPPT), which returns if a channel is PPT after a certain number of applications of the channel (as we didn't think we could implement a functions checking EPPT without an upper bound on the number of applications).

5.1.2 Experiment 1: Composition of Random Channels

We start by asking the question: for a random qubit channel Φ, how many times must we apply Φ before it becomes entanglement-breaking? Or more basic, are random qubit channels EPPT (and therefore EEB) with probability 1? We already expected to get a positive answer to the second questions because of the measure-zero results in [LG16]. To test this, we proceeded in the following way:

1. Generate 10^4 random qubit channels using `random_quantum_channel(2)`.
2. For each channel, compute its Choi matrix.
3. Use `is_eppt` to determine whether the composed channel Φ^n becomes PPT for $n = 1, 2, 3, \ldots$.
4. Count how many channels become EB at each iteration.

The key code snippet is shown below:

```
total = 10000
iterations = 0
count = 0

while count < total:
    count = 0
    for _ in range(total):
        chan = random_quantum_channel(2)
        choi = Choi(chan)
        if is_eppt(choi, iterations):
            count += 1
    print(f"EEB␣channels␣found␣after␣{iterations␣+␣1}␣
       applications:␣{count}/{total}")
    iterations += 1
```

Listing 5.2: Finding the EEB index of random qubit channels

The results confirm the theoretical rarity of entanglement-saving channels. We observe that:

- After 1 application: **2417/10000** channels are EB
- After 2 applications: **9319/10000**
- After 3 applications: **9981/10000**
- After 4 applications: **10000/10000**

This suggests that nearly all random qubit channels become EB after just a few iterations. This completely aligns with the results presented in the project, but we were surprised with the rapidness they become EB.

5.1.3 Experiment 2: Unitary Evolutions are Not EEB

As a control, we perform the same test on unitary channels generated with `random_unitary(2)` (another random generation function from Qiskit that

uses the Haar measure). Since unitary evolutions are known to be universally entanglement-preserving (UEP), as we saw in 3.2, they should never become EB regardless of how many times they are applied.

```
count = 0
total = 10000
iterations = 50
for _ in range(total):
    chan = random_unitary(2)
    choi = Choi(chan)
    if is_eppt(choi, iterations):
        count += 1

print(f"EEB unitary evolutions after {iterations} applications:
   {count}/{total}")
```

Listing 5.3: Test for EEB behavior of unitary channels

The result:

```
EEB unitary evolutions after 50 applications:  0/10000
```

confirms that no unitary channel becomes EB, even after 50 compositions, consistent with the theoretical result that unitary evolutions are ES.

5.2 Entanglement Saving

5.2.1 Experiment 3: Detecting Entanglement-Saving Channels

In the previous section, we explored the behavior of quantum channels under iteration, observing that almost all random qubit channels become entanglement-breaking after finitely many compositions. This suggests what we expected, that entanglement-saving (ES) channels are extremely rare.

To finally test this results, we wanted to implement numerical tests that determine whether a given qubit channel is entanglement-saving. To do so, we rely on the explicit classification of ES channels provided in theorem 3.3.1,

in which we will distinguish between unital and non-unital channels and characterize ES behavior via the Pauli Transfer Matrix (PTM) representation (see Lemma 2.3).

Implementation of the ES Check

In theorem 3.3, as we mentioned before, inversion of pure states refers to its geometric action in the Bloch sphere representation of qubit states. Recall that any qubit state ρ can be written as

$$\rho = \frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2},$$

where $\vec{n} \in \mathbb{R}^3$ with $|\vec{n}| \leq 1$, and $|\vec{n}| = 1$ for pure states. Then, a quantum channel ϕ *inverts* a pure state if

$$\phi\left(\frac{\mathbb{1} + \vec{n} \cdot \vec{\sigma}}{2}\right) = \frac{\mathbb{1} - \vec{n} \cdot \vec{\sigma}}{2}.$$

This is equivalent to a rotation of the Bloch vector $\vec{n} \mapsto -\vec{n}$. Importantly, this inversion property only makes sense for *unital* channels, as it requires the origin (the maximally mixed state) to remain fixed.

Then, we found it more convenient to implement the function differentiating between the unital and non-unital case.

In the unital case, which translates to $\vec{c} = 0$ in the PTM representation, these conditions translate directly to the existence of an eigenvalue of M with modulus 1. If $\lambda = \pm 1$, then the corresponding eigenvector $\vec{n}$ describes either a fixed or inverted pure state, satisfying the theorem.

In the non-unital case, we consider the image of a pure state $\rho = \frac{\mathbb{1}+\vec{n}\cdot\vec{\sigma}}{2}$ under ϕ, which becomes

$$\phi(\rho) = \frac{\mathbb{1} + (M\vec{n} + \vec{c}) \cdot \vec{\sigma}}{2}.$$

To say that $\phi(\rho) = \rho$ is then equivalent to solving

$$(M - \mathbb{1})\vec{n} = -\vec{c}, \quad \text{with} \quad \|\vec{n}\| = 1.$$

This is a constrained linear system with a norm constraint, and if a solution exists, then ϕ fixes a pure state.

Implementing the ES Check

The above analysis motivates the following function, which distinguishes between unital and non-unital cases and checks for the presence of fixed or inverted pure states:

- In the *unital case* ($\vec{c} = 0$), Φ is ES if and only if M has at least one eigenvalue of modulus one.
- In the *non-unital case*, Φ is ES if and only if the equation $(M - \mathbb{1})\vec{n} = -\vec{c}$ has a unit-norm solution $\|\vec{n}\| = 1$.

This logic is encoded in the following function:

```
def is_es_qubit(choi):
    ptm = PTM(SuperOp(choi)).data
    M = ptm[1:4, 1:4]
    c = ptm[1:4, 0]
    tol = 1e-7

    if np.isclose(np.linalg.det(M), 0, atol=tol):
        return False

    unital = np.allclose(c, np.zeros(3), atol=tol)

    if unital:
        eigenvalues = np.linalg.eigvals(M)
        return any(np.isclose(np.abs(eig), 1, atol=tol) for eig
            in eigenvalues)
    else:
        A = M - np.eye(3)
        b = -c
        n, residuals, rank, _ = np.linalg.lstsq(A, b, rcond=
            None)
        if residuals.size == 0 or np.allclose(residuals, 0,
            atol=tol):
            if np.isclose(np.linalg.norm(n), 1, atol=tol):
                return True
        return False
```

Listing 5.4: Checking if a qubit channel is Entanglement-Saving

5.2.2 Experiment 4: Testing ES on Unitary Evolutions

As a sanity check, we test the function `is_es_qubit` on randomly generated unitary channels (using the function `random_unitary(2)` as before). Since unitary channels are, by definition, universally entanglement-preserving (UEP), and hence entanglement-saving (ES), the function should return `True` in all cases.

```
count = 0
total = 1000
for _ in range(total):
    unitary = random_unitary(2)
    op_unitary = Operator(unitary)
    choi_unitary = Choi(op_unitary)
    if is_es_qubit(choi_unitary):
        count += 1

print(f"The␣number␣of␣ES␣unitary␣evolutions␣is:␣{count}/{total}
    ")
```

Listing 5.5: Verifying that all unitary channels are ES

The result confirms expectations:

```
The number of ES unitary evolutions is:  1000/1000
```

5.2.3 Experiment 5: Absence of Random ES Channels

Finally, we test whether any of the 10^4 random qubit channels generated via `random_quantum_channel(2)` are entanglement-saving. According to the theory, ES channels form a measure-zero subset of CPTP maps, so we expect to find none.

```
count = 0
total = 10000
for _ in range(total):
    chan = random_quantum_channel(2)
    choi = Choi(chan)
```

```
    if is_es_qubit(choi):
        count += 1

print(f"ES channels found: {count}/{total}")
```

Listing 5.6: Testing random channels for ES behavior

The experiment yields:

```
ES channels found:  0/10000
```

This strongly supports the theoretical result that generic channels are *not* entanglement-saving, as we expected from the previous experiments.

Code availability. All simulations and numerical experiments presented in this chapter were implemented in `Python` using `Qiskit`, and are openly available at the following GitHub repository:

`github.com/eescobarf/Saving-and-Breaking-Entanglement`

6 Appendix

Proof that the PPT criterion is sufficient in systems of size 2×2 and 2×3:

While it's easy to see that the PPT criterion is a necessary condition for a state to be separable, it's less straight forward to prove it is also sufficient for small sized systems (namely 2×2 and 2×3) [HHH96]. We will need the following

Lemma A.1
For any inseparable state $\tilde{\rho} \in \mathcal{H}_1 \otimes \mathcal{H}_2$, there exists a Hermitian operator $\tilde{A}$ such that

$$\mathrm{Tr}(\tilde{A}\tilde{\rho}) < 0 \quad \text{and} \quad \mathrm{Tr}(\tilde{A}\sigma) \geq 0$$

for any separable state σ.

Proof: Remember that the set of separable states is convex and closed in $\mathcal{H}_1 \otimes \mathcal{H}_2$. We can therefore apply a theorem which concludes from the Hahn-Banach theorem and states: If W_1, W_2 are convex closed sets in a real Banach space and one of them is compact, then there exists a continuous linear functional f and a real number $\alpha \in \mathbb{R}$ such that for all pairs $w_1 \in W_1$, $w_2 \in W_2$, we have

$$f(w_1) < \alpha \leq f(w_2).$$

Note that a one-element set consisting of $\tilde{\rho}$ is compact and the Hermitian operators on $B(\mathcal{H})_1 \otimes B(\mathcal{H})_2$ form a real-valued space $\tilde{A}$. It follows from the above (6) that there exists a real functional g such that

$$g(\tilde{\rho}) < \beta \leq g(\sigma)$$

As $\tilde{A}$ is a Hilbert space over the real numbers, g can be expressed as

$$g(\rho) = \mathrm{Tr}(\rho A)$$

with A being a Hermitian operator. Now as the trace of the states ρ, σ is 1 then it's clear that $\mathrm{Tr}(\beta I \rho) = \mathrm{Tr}(\beta I \sigma) = \beta$ with I meaning the identity. Due to the linearity of the trace we can then set the operator in the linear functional to

$$\tilde{A} = A - \beta I$$

to obtain the desired result. □

Using the above Lemma A.1, we can now prove an important theorem.

Theorem A.1
A state $\rho \in \mathcal{A}_1 \otimes \mathcal{A}_2$ is separable if and only if

$$\mathrm{Tr}(A\rho) \geq 0$$

for any Hermitian operator A satisfying $\mathrm{Tr}(AP \otimes Q) \geq 0$, where P and Q are projections acting on $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively.

Proof: The assuming ρ is separable, we can write is as $\rho = \sum_i p_i P_i \otimes Q_i$ where P_i, Q_i are positive and have trace equal to 1 and $\sum_i p_i = 1$. As $\mathrm{Tr}(AP_i \otimes Q_i) \geq 0$, it follows that $\mathrm{Tr}(A\rho) \geq 0$. To prove the other direction we will use the Lemma 6 from above and make a proof via contraposition. Assume ρ is not separable, then there exists a Hermitian operator A such that $\mathrm{Tr}(A\rho) < 0$ and $\mathrm{Tr}(A\sigma) \geq 0$ for any separable state σ, giving a contradiction.

This theorem deserves additional attention. The Hermitian operator A with the properties as described in the previous sentence, allows one to differentiate between the set of separable and inseparable states. It acts as a hyperplane and bears the intriguing name *entanglement witness*. It's important to point out that such a witness doesn't act as a witness for all inseparable states and at the same time, there is not a unique witness for any inseparable state.

The next step on our path to prove that PPT is sufficient in systems 2×2 and 2×3, we prove a slightly modified version of theorem 4.1.

Theorem A.2
Let ρ act on the Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$. Then ρ is separable if and only if for any positive map $\Lambda : \mathcal{H}_2 \to \mathcal{H}_1$, the operator $(I \otimes \Lambda)\rho$ is positive.

If ρ is separable, it is clear that the condition holds. To prove the other direction, we remind ourselves of the Jamiolkowski's theorem that says

$$\begin{aligned} & \Lambda \in \mathcal{L}(\mathcal{H}_1, \mathcal{H}_2) \quad \text{is positive} \\ \iff \quad & \mathcal{J}(\Lambda) = \mathcal{J}(\Lambda)^\dagger \quad \text{and} \quad \operatorname{Tr}\Big(\mathcal{J}(\Lambda)\, P \otimes Q\Big) \geq 0 \\ & \text{for all projectors } P \in \mathcal{A}_1,\ Q \in \mathcal{A}_2 \end{aligned}$$

with $\mathcal{J}(\Lambda) = \sum_{i,j=1}^{n} E_{ij} \otimes \Lambda(E_{ji})$. As the transposition map T is positive and $T^2 = I$, it follows that any positive map Λ can be written as a composition of T and another positive map (specifically $\Lambda \circ T$). This brings us to an equivalent statement of the Jamiolkowski's theorem but in terms of the Choi state.

$$\begin{aligned} & \Lambda \in \mathcal{L}(\mathcal{A}_1, \mathcal{A}_2) \quad \text{is positive} \\ \iff \quad & \mathcal{C}(\Lambda) = \mathcal{C}(\Lambda)^\dagger \quad \text{and} \quad \operatorname{Tr}\Big(\mathcal{C}(\Lambda)\, P \otimes Q\Big) \geq 0 \\ & \text{for all projectors } P \in \mathcal{A}_1,\ Q \in \mathcal{A}_2 \end{aligned}$$

with $\mathcal{C}(\Lambda) = \sum_{i,j=1}^{n} E_{ij} \otimes \Lambda(E_{ij})$. We now identify a Hermitian operator as a Choi matrix with a corresponding map. Comparing with theorem A.1, gives us the following expression

$$\operatorname{Tr}\Big([(I \otimes \Lambda)\, w]\rho\Big) \geq 0$$

where ω denotes the corresponding matrix of the unnormalized maximally entangled state $\sum_{i,j=1}^{n} E_{ij} \otimes E_{ij}$. The adjoint of a positive map is also positive. Therefore, the statement below is equivalent

$$\operatorname{Tr}\Big(w\,(I \otimes \Lambda)\rho\Big) \geq 0$$

for any positive map $\Lambda : \mathcal{H}_2 \to \mathcal{H}_1$. We have successfully reformulated the theorem in terms of positive maps and interestingly, the previously the

restriction for Hermitian operators $\mathrm{Tr}(AP \otimes Q) \geq 0$, where P and Q are projections acting on $\mathcal{H}_1$ and $\mathcal{H}_2$ from theorem A.1 is not needed anymore as it is trivially fullfilled. As ω is a one-dimensional projector, we can rewrite the expression further

$$\langle\Omega|\,(I \otimes \Lambda)\rho\,|\Omega\rangle \geq 0$$

with $|\Omega\rangle = \sum_{i=1}^{n} |i,i\rangle$ denoting the maximally entangled state. Finally we conclude, if $(I \otimes \Lambda)\rho$ is positive for any positive map $\Lambda : \mathcal{A}_2 \to \mathcal{A}_1$ then the theorem is fullfilled and the ρ is separable. □

Putting everything together, we now have the tools to prove that the PPT criterion fully characterises the states acting on $\mathbb{C}^2 \otimes \mathbb{C}^2$ and $\mathbb{C}^2 \otimes \mathbb{C}^3$. Strømer and Woronowicz showed that any positive map $\Lambda : \mathcal{A}_1 \to \mathcal{A}_2$ with $\mathcal{H}_1 = \mathcal{H}_2 = \mathbb{C}^2$, or $\mathcal{H}_1 = \mathbb{C}^3$, $\mathcal{H}_2 = \mathbb{C}^2$ (equivalently $\mathcal{H}_1 = \mathbb{C}^2$, $\mathcal{H}_2 = \mathbb{C}^3$), is decomposable i.e. of the form

$$\Lambda = \Lambda_1^{\mathrm{CP}} + \Lambda_2^{\mathrm{CP}} \circ T, \tag{6.1}$$

where Λ_i^{CP} are completely positive maps, and T denotes the transpose map. It follows that if $\rho^{T_2} \geq 0$ then $(I \otimes \Lambda_1^{\mathrm{CP}})\rho + (I \otimes \Lambda_2^{\mathrm{CP}})\rho^{T_2}$ is also positive. Therefore theorem A.2 holds and the PPT criterion is also sufficient for these cases. □

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